

Applied Statistics (MATH10096)

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Practical modelling with factors and interactions

A simple example

- As an example of the use of factor variables, consider the `PlantGrowth` data frame supplied with R.
- These are data from an experiment to compare yields (as measured by dried weight of plants) obtained under a control and two different treatment conditions.
- The factor `group` has three levels: `ctrl`, `trt1` and `trt2`, and it is believed that the growth of the plants depends on this factor.
- First check that `group` is already a factor variable:

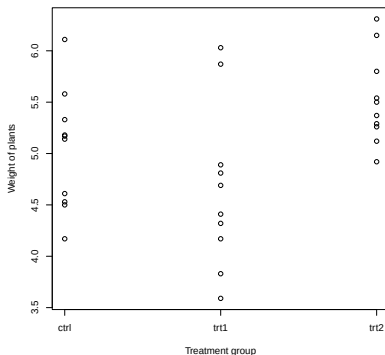
```
PlantGrowth$group
## [1] ctrl ctrl ctrl ctrl ctrl ctrl ctrl ctrl ctrl ctrl trt1 trt1 trt1 trt1 trt1
## [16] trt1 trt1 trt1 trt1 trt1 trt1 trt2 trt2 trt2 trt2 trt2 trt2 trt2 trt2 trt2 trt2
## Levels: ctrl trt1 trt2
is.factor(PlantGrowth$group)
## [1] TRUE
```

- The aim is to investigate whether the `group` factor controls the weight of the plant, and if so, to what extent.

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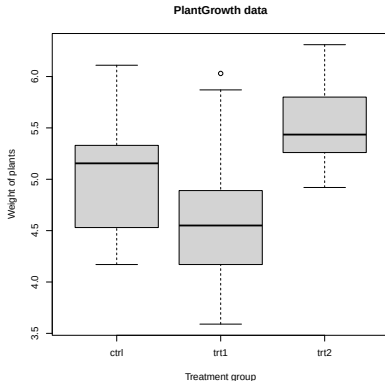
```
stripchart(weight ~ group, data = PlantGrowth, vertical = TRUE, pch = 1,  
           ylab = "Weight of plants", xlab = "Treatment group")
```



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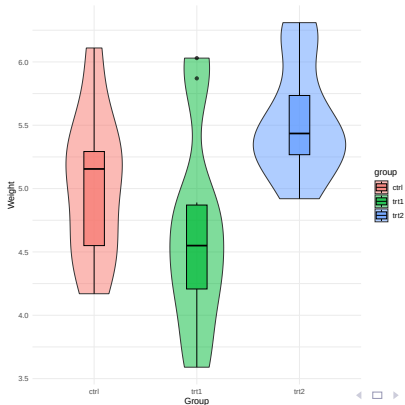
```
boxplot(weight ~ group, data = PlantGrowth, main = "PlantGrowth data",  
        ylab = "Weight of plants", xlab = "Treatment group")
```



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```
require(ggplot2)
ggplot(PlantGrowth, aes(x = group, y = weight, fill = group)) +
  geom_violin(alpha = 0.5) +
  geom_boxplot(width = 0.2, color = "black", alpha = 0.7) + # Boxplot inside
  theme_minimal() +
  labs(x = "Group", y = "Weight")
```



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- ↪ Treatment 2 seems to stand out in terms of the corresponding weight of the plants.
- ↪ Let us fit the model now.

```
pgm.1 <- lm(weight ~ group, data = PlantGrowth)
head(model.matrix(pgm.1))

##      (Intercept) grouptrt1 grouptrt2
## 1             1         0         0
## 2             1         0         0
## 3             1         0         0
## 4             1         0         0
## 5             1         0         0
## 6             1         0         0

contrasts(PlantGrowth$group)

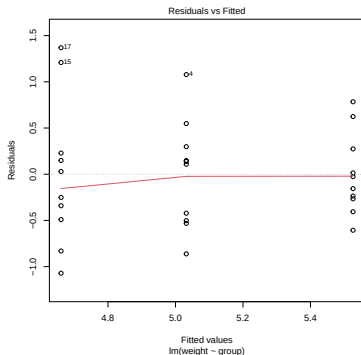
##      trt1 trt2
## ctrl   0   0
## trt1    1   0
## trt2    0   1
```

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→ First thing to do, after fitting a model, is to check the residual plots.

```
plot(pgm.1, which = 1)
```

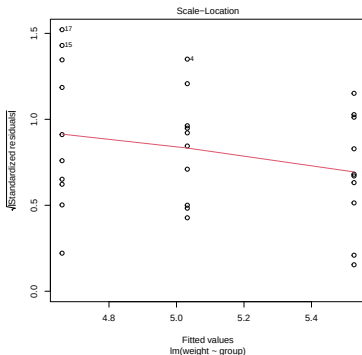


Practical modelling with factors and interactions

A simple example

→ First thing to do, after fitting a model, is to check the residual plots.

```
plot(pgm.1, which = 3)
```

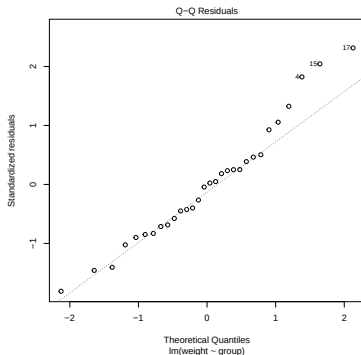


Practical modelling with factors and interactions

A simple example

→ First thing to do, after fitting a model, is to check the residual plots.

```
plot(pgm.1, which = 2)
```

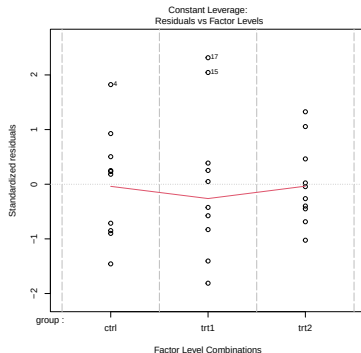


Practical modelling with factors and interactions

A simple example

→ First thing to do, after fitting a model, is to check the residual plots.

```
plot(pgm.1, which = 5)
```



Practical modelling with factors and interactions

A simple example

→ Let us now inspect the summary output.

```
summary(pgm.1)
##
## Call:
## lm(formula = weight ~ group, data = PlantGrowth)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.0710 -0.4180 -0.0060  0.2627  1.3690
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.0320     0.1971  25.527  <2e-16 ***
## grouptrt1     -0.3710     0.2788  -1.331   0.1944
## grouptrt2      0.4940     0.2788   1.772   0.0877 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6234 on 27 degrees of freedom
## Multiple R-squared:  0.2641, Adjusted R-squared:  0.2096
## F-statistic: 4.846 on 2 and 27 DF,  p-value: 0.01591
```

→ The estimated overall mean weight (in the population that these plants represent, given **control conditions**) is 5.032, while treatment 1 is estimated to lower this weight by 0.37, and treatment 2 to increase it by 0.49.

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- ↪ Neither parameter individually appears to be significantly different from zero.
- ↪ Usually, it is only of interest to test whether the whole factor variable should be in the model or not, and this amounts to testing whether all its associated parameters are simultaneously zero or not.
- ↪ We can do this with an F -ratio test and because the only predictor in this model is the factor variable `group`, this corresponds exactly to the F statistic reported in the very last line of the summary output.
- ↪ The p-value is 0.016 suggesting that the null hypothesis that the group variable does not have an effect on weight is not very plausible.
- ↪ So we see that the data provide evidence for an effect of the `group` factor variable on `weight`, which appeared marginal or absent when we examined p-values for the individual model parameters.

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- Let us now create a new group variable and change the baseline level from the control group to the treatment 1 group.

```
PlantGrowth$group_alt <- relevel(PlantGrowth$group, ref = "trt1")
pgm.l_alt <- lm(weight ~ group_alt, data = PlantGrowth)
summary(pgm.l_alt)

##
## Call:
## lm(formula = weight ~ group_alt, data = PlantGrowth)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.0710 -0.4180 -0.0060  0.2627  1.3690
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.6610     0.1971  23.644 < 2e-16 ***
## group_altctrl    0.3710     0.2788   1.331  0.19439
## group_altttrt2    0.8650     0.2788   3.103  0.00446 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6234 on 27 degrees of freedom
## Multiple R-squared:  0.2641, Adjusted R-squared:  0.2096
## F-statistic: 4.846 on 2 and 27 DF, p-value: 0.01591
```

Practical modelling with factors and interactions

A simple example

- ↪ Aside from the coefficients section, the rest of the output remains exactly the same.
- ↪ Note that under the first model (in which the control group is the baseline) the estimated mean plant growth in the control group is 5.0320, in the treatment 1 group is $5.0320 - 0.3710 = 4.661$ and in the treatment group 2 is $5.0320 + 0.4940 = 5.526$.
- ↪ In turn, in the second model (in which the treatment 1 group is the baseline), the estimated mean plant growth in the control group is $4.6610 + 0.3710 = 5.032$, in the first treatment group is 4.6610 and in the second treatment group is $4.6610 + 0.8650 = 5.526$.
- ↪ Further, and for instance, under the first model, the estimated difference in mean plant growth between treatment group 1 and 2 is $-0.3710 - 0.4940 = -0.865$, whereas such difference in the second model is given by -0.8650 .